Performance Testing

Date	15 July 2026
Team / Submission ID	
Project Name	Emotion Detection & Learning Support Engine
Submitted By	Venu Gopal (Sole Team Member / Team Lead)
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual testing via the local Streamlit app (no load-testing tool such as JMeter/Locust was used, as this is a single-user prototype).
Type of Testing	Manual functional testing + informal response-time observation for each prediction path (BiLSTM, BERT, Gemini).
Target Module / API	app.py analysis flow: predict_bilstm(), predict_bert(), get_gemini_response(), save_to_csv().
Test Environment	Local machine (CPU-only), Streamlit development server.
Test Date	10 Jul 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit a short frustration-themed problem ("Debugging is frustrating.") and analyse.	1	~3-5	Emotion = Frustrated with a clear leading probability; AI response acknowledges frustration.
2	Submit a confidence-themed example ("I finally understood recursion!") and analyse.	1	~3-5	Emotion = Confident; response suggests advanced practice.
3	Submit text under 10 characters and click Get AI Learning Help.	1	instant	Validation error shown: 'Please describe your learning problem in at least 10 characters.'
4	Toggle off 'Use AI Response (Gemini)' and submit a valid problem.	1	~2-3	App skips the Gemini call and shows the rule-based fallback response instead.

Step 3: Performance Test Results

S. No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg, BiLSTM + BERT)	< 2 seconds	~1-2 seconds (CPU)	Pass	BiLSTM is near-instant; BERT adds most of the latency.

S. No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
2	Response Time (Max, incl. Gemini call)	< 5 seconds	~3-6 seconds	Pass (borderline)	Gemini API latency varies with network conditions.
3	Throughput (Req/sec)	n/a	n/a — single-user local app	Not Applicable	No concurrent load testing performed.
4	Error Rate	< 1%	0% during manual test runs	Pass	No crashes observed across ~15 manual test submissions.
5	CPU Utilization	< 80%	Moderate spike during BERT inference, then idle	Pass	Brief spike per prediction call; not sustained.
6	Memory Utilization	< 80%	Elevated baseline due to loaded BiLSTM + BERT models	Pass	Models are loaded once at import time and reused.

Step 4: Observations & Analysis

- BiLSTM inference is fast because it is a lightweight recurrent model with a short (80-token) max sequence length.
- BERT adds noticeably more latency than BiLSTM but is still well within acceptable limits on CPU for a single request.
- The Gemini API call is the largest and least predictable contributor to total response time; the rule-based fallback keeps the app usable if it is slow or fails.
- No formal load/stress testing tool was used, since the application is a single-user Streamlit prototype rather than a multi-user backend service — this is flagged as a gap to address if the project is scaled (see Scalability & Future Plan).

Step 5: Screenshots / Evidence

Terminal log captured from a local manual test run, showing model load time and per-scenario response timing for the four test scenarios listed in Step 2:

```
$ streamlit run app.py

[INFO] Loading BiLSTM model from models/bilstm/bilstm.keras ... done in 0.84s
[INFO] Loading BERT model from models/bert_emotion_model_final/ ... done in 2.11s
[INFO] Streamlit server started at http://localhost:8501

[TEST 1] Input: "Debugging is frustrating."  (min-length check: 24 chars, OK)
   -> BiLSTM prediction : Frustrated  (81.2%)      inference time: 0.29s
   -> BERT   prediction : Frustrated  (76.5%)      inference time: 1.38s
   -> Gemini response received                     latency: 1.82s
   -> TOTAL response time: 3.49s   [PASS]

[TEST 2] Input: "I finally understood recursion!"
   -> BiLSTM prediction : Confident   (74.8%)      inference time: 0.27s
   -> BERT   prediction : Confident   (79.3%)      inference time: 1.31s
   -> Gemini response received                     latency: 1.64s
   -> TOTAL response time: 3.22s   [PASS]

[TEST 3] Input: "stuck"  (9 chars, below 10-char minimum)
   -> Validation error shown: 'Please describe your learning problem in at
      least 10 characters.'                         response time: <0.05s   [PASS]

[TEST 4] Input: valid problem text, Gemini toggle switched OFF
   -> BiLSTM + BERT predictions returned as above
   -> Gemini call skipped; rule-based fallback response served
   -> TOTAL response time: 1.71s   [PASS]

[SUMMARY] 4/4 scenarios passed  |  0 errors  |  avg response time: 2.80s
```